**Department of Computer Science and Applications**
**Panjab University, Chandigarh**

## 15-days Skill Enhancement Course/Workshop

## On

## "Introduction to Data Science"

## Report

The course was organized by Department of Computer Science and Applications under RashtriyaUchchatarShikshaAbhiyan (RUSA) project for skill enhancement courses in Computer Science and Applications. The course coordinators of the workshop: Dr.Rohini Sharma, Chairperson and Ms. Jasleen Kaur Bains along with Prof.Sonal Chawla, Principle Investigator, RUSA project for Skill enhancement courses coordinated this workshop. This course was started as per the mandate of National Education Policy as a step towards inculcating skill based education among masses as data Science is an upcoming field and people from various areas are using data science concepts in their work. For this course more than 300 registrationsfrom different states of the country like, chandigarh, Punjab, Haryana, himachalpradesh, rajasthan, Karnataka etc. were received. More than 100 participants have attended this workshop sincerely attending more that 75% of the sessions. This 30 hours workshop was conducted in online mode from 4:00 pm to 6:00 pm everyday so that the students and working professionals could attend.

In the very first session of the workshop, Prof. R.K. Singla talked about Data Science and the need for its awareness. He discussed the distinction between the artificial intelligence, machine learning and deep learning in depth and demonstrate different machine learning models using Google Teachable Machine and Lobe.AI.

Dr.Rohini Sharma took second session and explained the foundational concepts of Python Programming and demonstrate the use of Jupyter notebook with use cases and hands on practice as python is the most popular language among data scientists because of its rich features.

Ms. Supreet Kaur Mann, Assistant Professor in the department discussed object oriented concepts and their implementation in python. She further explained the libraries which are available for use in python.

Dr.Anuj Kumar, Assistant Professor in the department discussed data visualisation library along with other libraries like numpy, pandas etc. He explained how different libraries and functions are used in python programming for Data visualization.

Having covered the core concepts and some of the use-cases of Python programming, it was the perfect time to move to the domain of machine learning and its applications. Prof.Anu Gupta took the session on machine learning and its applications. She emphasized the importance of machine learning and how everything is getting in automation these days. Mr. AbhinavKathuria working as assistant professor in RSD College Ferozepur further gave practical demonstration of few of its applications.

The next talk on Data Regression was delivered by Prof. Pradeep Kumar, from Kurukshetra University. The core concepts were beautifully explained and further he gave practical Demonstration of Data Regression techniques using Python programming.

Dr.Anuj Sharma, Assistant Professor in the department delivered a talk on introduction to Data Classification and techniques. With a pictorial representation, he illustrated how different fields emerge from AI. The concept of Reinforcement and Transfer learning was also explained from mathematical point of view. Mr. Ujjwal, working as Assistant Professor on Guest basis demonstrated the concepts in python.

Ms. Supreet Kaur took another session on Data Clustering and its techniques which is her domain of expertise. She explained different clustering techniques with practical demonstration.

Prof.Indu Chhabra delivered talk on Deep learning. She explained the difference of Supervised Learning and unsupervised learning with delightful examples. Prof. Chhabra related the concepts of deep learning with every day's life.Ms. Samita, research scholar in the department further demonstrated the implementation of the Artificial Neural network using Python programming.

Dr.Anuj Sharma took an enlightening session on Deep learning techniques. He explained the working of hidden layers, activation functions, inputs and outputs. He explained the concept of Neural Networks and how the results could be optimized by changing weights during back propagation. Further he discussed Convolution Neural Network.

Dr.UshaBhadera, Assistant Professor, Jaipuria Institute of Management took the session on Data Visualization. She demonstrated data visualization and its use-cases using the software Tableau Public. The software was indeed of great help as it was easy to use and removed the barriers in Microsoft Excel.

Dr.KavitaTaneja, Assistant Professor in the departmentdelivered a talk on Introduction to Natural Language Processing. Dr.Taneja apprised the participants with the recent trends of ChatGPT, bard etc. She explained the steps in Natural Language Processing with practical implementation.

After gaining enough knowledge about data science concepts, the techniques and practical implementation, a session on Research Data Analytics and Managing Research Data was taken by Dr.Neeraj Kumar Singh, Deputy Librarian, Panjab University. He highlighted the importance of correct sources of information for the Research Scholars and how analysis of Data should be done. He also underlined the need for managing Research data in order to alleviate the problems Research Scholars face during their research.

The participants applauded this workshop and requested to organized more such workshops on different topics.

In this workshop, many people are involved in one way or the other to organize it successfully. The faculty members of the department were whole heartedly involved in this workshop as most of the sessions have been taken by our esteemed and knowledgeable faculty members only. The technical and office staff played their part really well as no technical glitch has been reported throughout the workshop. The external support provided by computer centre is also well appreciable.The research scholars specially who were part of the organizing team worked really hard to make this workshop a success: Arun Sharma, Vaishali, Lalita Thakur and Ujjwal Thakur. The excellent feedback received for this workshop motivated the department to organize more such workshops.

**About the University**
One of the oldest universities in India, the Panjab university (PU) initiated at Lahore in 1882, has a long tradition of pursuing excellence in teaching and research in science and technology, humanities, social sciences, performing arts and sports. The university supports excellence and innovation in academic programs, promotes excellence in research, scholarship and teaching. The university is committed to attract and support the best students and faculty, who excel at teaching and research. In Independent India, Panjab University with its Campus at Chandigarh and nearly 188 colleges in Panjab state and Chandigarh U.T., has served various societal needs with distinction.

**About the Department**
The Computer Culture at the Panjab University dates back to 1966. An independent Centre for Computer Science and Applications (now a full fledged department) was set-up in 1983. The department aims at ingraining the spirit of ingenuity, innovativeness and technical competence in the students through rigorous competition and regular guidance. The department also caters to the need of users in the region and organizes programs for teaching and research communities.

**About the Course**
The proposed 15 days course/workshop is focused to impart foundational level knowledge in the field of Data Science to a big chunk of students ranging from undergraduate level to post graduate level and even to research scholars and faculty at large. The course pertains to one of the most evolving domain of present era and participants from various disciplines may be benefitted in their respective research and academic work.

**Duration : 15 Days**
**Mode : Online**
**Timings : 4 pm to 6 pm**
**Dates :1st - 21st February 2023,**
**(Monday – Friday)**
**Registration Link:**
**https://forms.gle/8AFkPqX71ccd3sE96**
**Last date to register is 30-01-2023 till 5:00pm**
**(The link of the workshop will be shared with registered participants selected on First Come First Serve basis)**

**Chief Patron**

**Prof Renu Vig,**
**Hon'ble Vice Chancellor,**
**Panjab University, Chandigarh**

**Core Contents**

- Artificial Intelligence
- Machine Learning
- Data Regression
- Data Clustering
- Natural Language Processing (NLP)
- Python Programming
- Date Visualization
- Data Classification
- Deep Learning
- Future Scope of Data Science

**Speakers**

- Prof. Pradeep Kumar (KU)
- Prof. Indu Chhabra (DCSA PU)
- Prof. Sonal Chawla (DCSA PU)
- Dr. Anuj Sharma (DCSA PU)
- Dr. Kavita Taneja (DCSA PU)
- Prof. R.K. Singla (DCSA PU)
- Prof. Anu Gupta (DCSA PU)
- Dr. Rohini Sharma (DCSA PU)
- Dr. Anuj Kumar (DCSA PU)
- Ms. Supreet Kaur (DCSA PU)

**Dr. Rohini Sharma,**
**Chairperson, DCSA**
**Course Coordinator**

**Ms. Jasleen Kaur Bains,**
**Assistant Professor, DCSA**
**Course Coordinator**

**Prof. (Dr.) Sonal Chawla,**
**Principal Investigator, RUSA Project,**
**Skill Enhancement Courses**
**in**
**Computer Science and Applications**

**Department of Computer Science and Applications**
**Panjab University, Chandigarh.**

**Panjab University**
**Chandigarh**

**A SKILL ENHANCEMENT COURSE/ WORKSHOP**
**ON**
**INTRODUCTION TO DATA SCIENCE**
**(in online mode)**

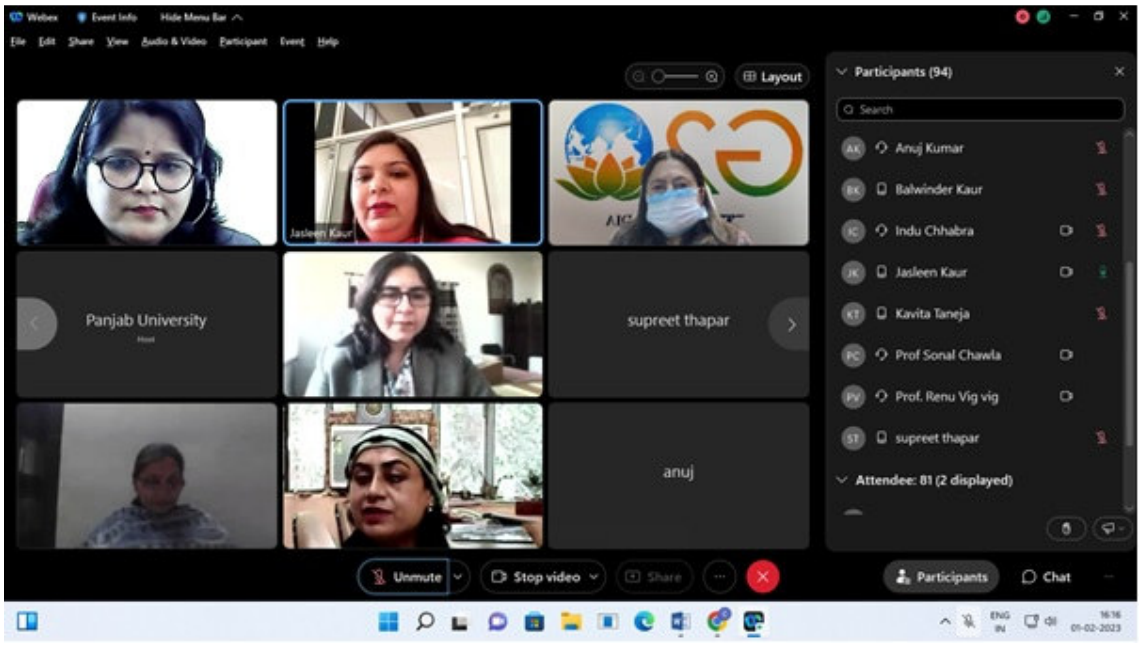
**1st - 21st February, 2023**

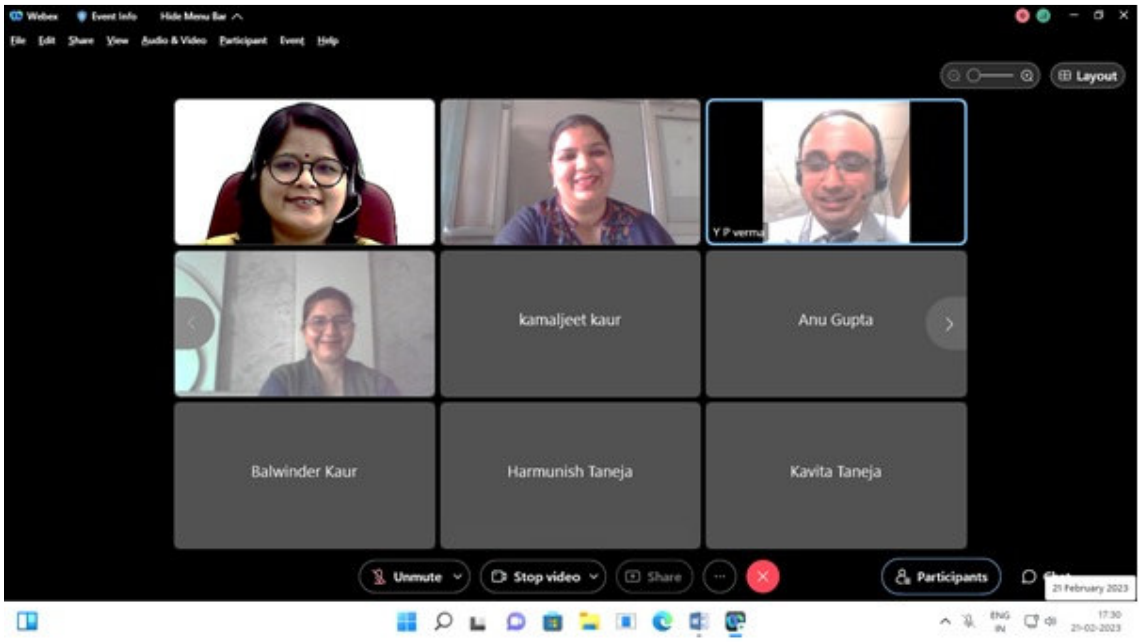
Organized by
**Department of Computer Science & Applications (DCSA), Panjab University, Chandigarh, India**

*in collaboration with*
**Rashtriya Uchchatar Shiksha Abhiyan (RUSA)**



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Y P verma
kamaljeet kaur
Anu Gupta
Balwinder Kaur
Harmunish Taneja
Kavita Taneja
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PUNJAB EXPRESS

# 15-day Skill Enhancement Course and Workshop on "Introduction to Data Science" begins at PU

**PUNJAB EXPRESS BUREAU**
Chandigarh, February 2

The 15-days Skill Enhancement Course & Workshop on "Introduction to Data Science" organized by Department of Computer Science and Applications, Panjab University, Chandigarh began here on Thursday.

Dr. Rohini Sharma, Chairperson and Coordinator of the program shared about the department and the importance of the workshop according to National Education Policy. More than 300 participants from different states registered and joined the program. Prof. Sonal Chawla, Principal Investigator RUSA project for skill enhancement programmes in computer science and applications shared the vision of Rashtriya Uchchtar Shiksha Abhiyan. Prof. Renu Vig, VC inaugurated the programme and stressed upon the importance of Data Science and need of more such workshops with flexible timings of this kind which could benefit the participants and further help the purpose of the National Education Policy.

# इंट्रोडक्शन टू डाटा साइंस पर स्किल अनाउंसमेंट कोर्स शुरू

चंडीगढ़ | पंजाब यूनिवर्सिटी में इंट्रोडक्शन टू डाटा साइंस पर डिपार्टमेंट ऑफ कम्प्यूटर साइंस एंड एप्लीकेशन की ओर से 15 दिन के कोर्स व वर्कशॉप की शुरुआत वीरवार को हुई, जिसका उद्घाटन वीसी प्रो. रेनू विग ने किया। राष्ट्रीय उच्चतर शिक्षा अभियान के तहत हो रही वर्कशॉप में करीब 300 पार्टिसिपेंट्स ने रजिस्टर्ड किया था। मुख्य वक्ता रहे पूर्व डीयूआई प्रो. आरके सिंगला। प्रो. सिंगला ने कहा कि डाटा एनालिटिकल के बारे में अवेयरनेस की बहुत जरूरत है। आने वाले दिनों में यह सबसे बड़ा जॉब सेक्टर होगा, इसलिए जरूरी है कि इसकी हैंड आन ट्रेनिंग भी दी जाए।

**AMAR UJALA**

# विद्यार्थियों को बताया डाटा साइंस का महत्व

**चंडीगढ़।** पंजाब यूनिवर्सिटी के कंप्यूटर विज्ञान और अनुप्रयोग विभाग की ओर से डाटा साइंस का परिचय पर 15 दिवसीय कौशल वृद्धि पाठ्यक्रम का आयोजन किया गया। कार्यक्रम की अध्यक्ष एवं समन्वयक डॉ. रोहिणा शर्मा ने राष्ट्रीय शिक्षा नीति के अनुसार कार्यशाला का महत्व बताया।

इसमें 300 से अधिक प्रतिभागियों ने पंजीकरण करवाया। पूर्व डीयूआई प्रो. आरके सिंगला ने बताया कि निकट भविष्य में सबसे अधिक मांग डाटा साइंस क्षेत्र में होगी। संवाद

PUNJAB EXPRESS

# Over 300 attend fortnight-long workshop at PU

PUNJAB EXPRESS BUREAU
Chandigarh, February 22

A fortnight-long Skill Enhancement Course/Workshop under the Rashtriya Uchchatar Shiksha Abhiyan (RUSA) project concluded on Tuesday with the presence of Dr Yajvender Pal Verma, Registrar, Punjab University who acknowledged the need of such events and shared his experience among the participants regarding the emergent need of machine learning and its principles for various domains. He stressed on interdisciplinary courses and research using machine learning. He expressed his willingness to join such future workshops.

The course Coordinators Dr. Rohini Sharma, Chairperson and Jasleen Kaur Bains apprised about the course where more than 300 participants registered from different parts of the region. Due to the online nature and evening timing, large number of students, working professionals, research scholars and faculty members could attend the workshop.